

Hardware Specifications & Prototyping Report

Author: James Guentert • Technical Documentation Suite

This technical report documents the analog front-end stage of the Electrocardiogram (ECG) instrumentation amplifier system. The design bridges physical biopotential extraction with a National Instruments ELVIS II workstation for software-based digital signal processing pipelines.

1. Three-Stage Instrumentation Amplifier Schematic & Parameters

The circuit layout uses a classic three-stage operational amplifier configuration optimized for extracting microvolt-level cardiac biopotentials while maintaining high input impedance and robust common-mode noise rejection. Below are the empirically evaluated system parameters and corresponding circuit diagram.

Parameter	Measured Value	Design Significance
Dynamic Range	29.6 mV	Accommodates raw physiological differential spikes without input saturation.
DC Offset Voltage	1.1 V	Compensated via a downstream high-pass filtering stage in software or hardware.
Frequency Response	0.1 Hz - 80 Hz	Tuned precisely to capture clinical QRS complexes while rejecting out-of-band noise.
Differential Gain	53.33 dB	Provides the requisite voltage amplification to bring microvolt traces to line-level DAQ scale.
Common-Mode Rejection Ratio (CMRR)	68.45 dB	Ensures clean signal output by aggressively suppressing ambient 60 Hz powerline hum.

Dynamic range	DC offset voltage	Frequency response range	Differential gain	CMRR
29.6 mV	1.1V	0.1 Hz - 80 Hz	53.33 dB	68.45 dB

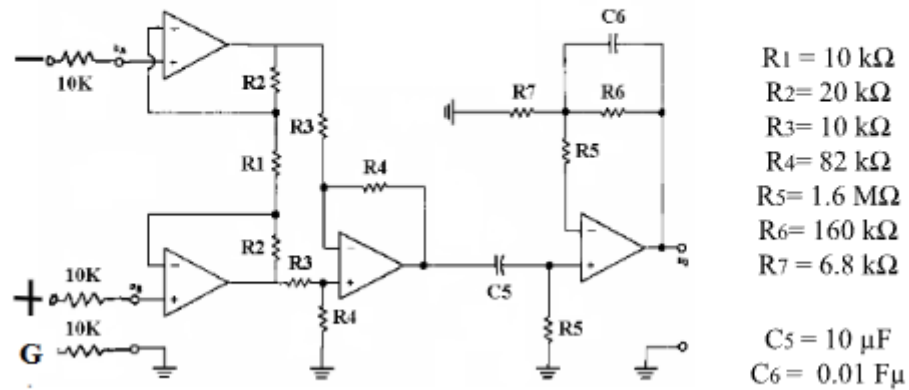


Figure 3A: Three-stage instrumentation amplifier schematic

Figure 1: Complete circuit topology including input protection resistors, pre-amplifier staging, differential amplification, and high-pass/low-pass analog filtering components.

2. Physical Benchtop Prototyping & Workspace Setup

The physical implementation was deployed on an NI ELVIS II Series Prototyping Board. Resistor and capacitor arrays were manually integrated alongside multi-channel operational amplifier integrated circuits. Solid-core jumper wires manage connections between the analog front-end, active power supply rails (+15V / -15V), analog ground networks, and the workstation's analog input (AI) channel interfaces.

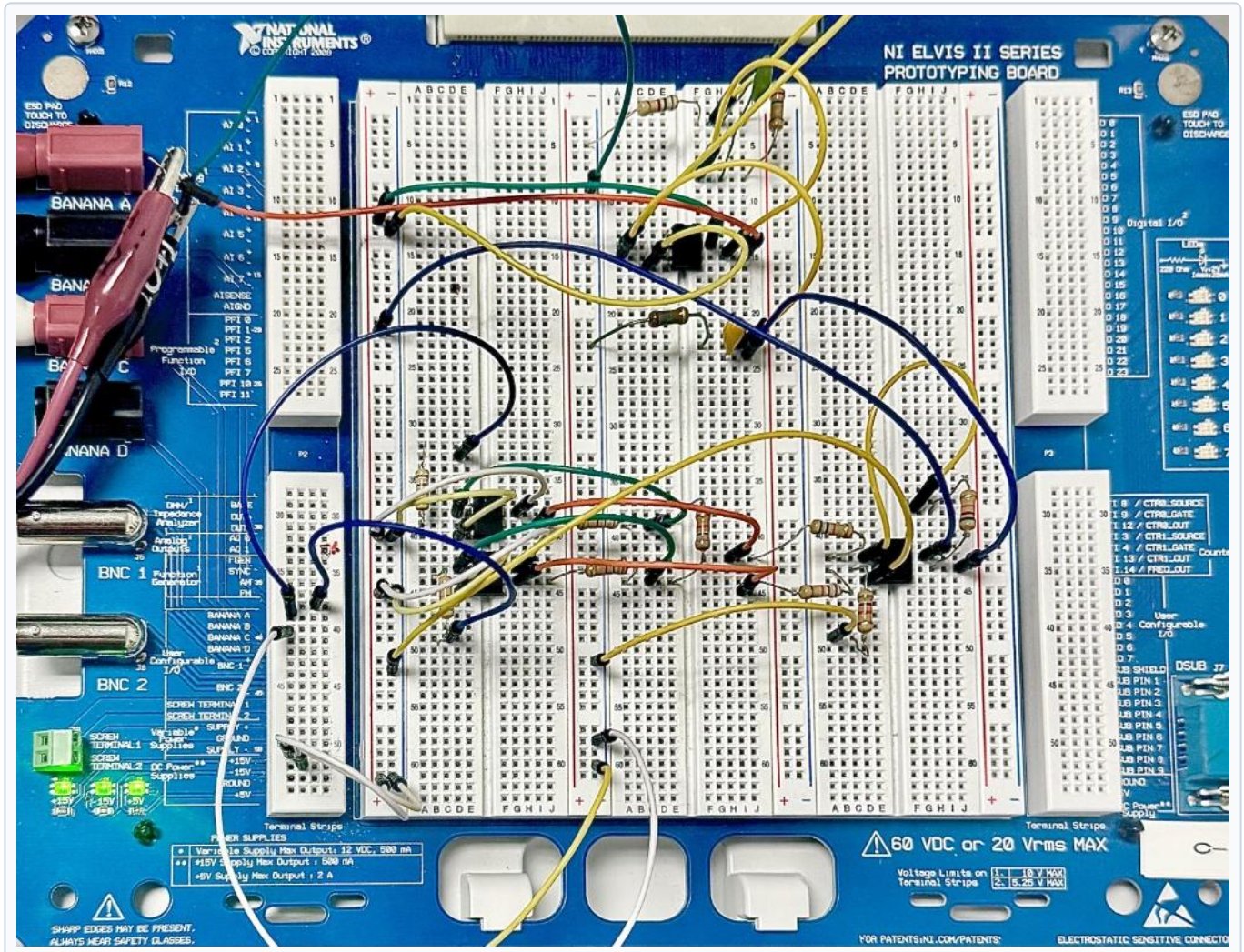


Figure 2: Physical breadboard assembly showcasing precision component placement, multi-stage layout separation, and shielded cable lines linked to the validation hardware.

3. Structural Hardware Interfacing

The physical layout directly implements the resistance profiles specified in the schematic: R1 (10 k Ω), R2 (20 k Ω), R3 (10 k Ω), R4 (82 k Ω), R5 (1.6 M Ω), R6 (160 k Ω), and R7 (6.8 k Ω), integrated alongside decoupling and filtering capacitors C5 (10 μ F) and C6 (0.01 μ F). This distribution stabilizes the pre-amplifier inputs, matching bodily electrode contact impedances and maintaining safe isolation constraints before passing structural telemetry into the MATLAB Data Acquisition Toolbox suite.